

MATH 2270 — Linear Algebra

Practice Test 3 — Eigenvalues, Inner Products & Orthogonality

Unit 3: §5.1–5.3, 5.5 (Eigenvalues, Diagonalization, Markov Chains) and §6.1–6.5, 7.1 (Inner Products, Gram–Schmidt, Least Squares, Orthogonal Matrices)

Name: _____

Date: _____

Practice test — twice as long as the real test (200 points).

The real Test 3 is 100 points in 50 minutes, so allow about 100 minutes for this whole practice test, or work through half of each section in 50 minutes. None of these questions appear on the real test. Same rules as the real test: closed book, closed notes, and a row-reduction calculator or software is allowed for Section 6 ONLY.

Section 1 — True / False

36 pointsCircle **T** or **F** for each statement.

- | | |
|--|--------------|
| 1. If A is a triangular matrix, then its eigenvalues are its diagonal entries. | T / F |
| 2. The zero vector can be an eigenvector of a matrix. | T / F |
| 3. Every orthogonal set of nonzero vectors in an inner product space is linearly independent. | T / F |
| 4. If a square matrix is diagonalizable, then it is invertible. | T / F |
| 5. If an $n \times n$ matrix has n distinct eigenvalues, then it is diagonalizable. | T / F |
| 6. The rotation of $\mathbb{R}^2$ by 90° has no real eigenvalues. | T / F |
| 7. Eigenvectors belonging to different eigenvalues of a matrix are linearly independent. | T / F |
| 8. If A and B are similar, then A and B have the same eigenvectors. | T / F |
| 9. For a regular Markov chain, the steady-state vector does not depend on the initial state vector $\mathbf{x}_0$. | T / F |
| 10. In every real inner product space, $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$. | T / F |
| 11. In every inner product space, $\ \mathbf{u} + \mathbf{v}\ = \ \mathbf{u}\ + \ \mathbf{v}\ $. | T / F |
| 12. In every inner product space, $ \langle \mathbf{u}, \mathbf{v} \rangle \leq \ \mathbf{u}\ \ \mathbf{v}\ $. | T / F |
| 13. For a subspace W , the only vector in both W and $W^\perp$ is $\mathbf{0}$. | T / F |
| 14. In Gram–Schmidt, the second vector $\mathbf{u}_2$ is orthogonal to $\mathbf{v}_1$. | T / F |
| 15. If A has linearly independent columns and $A = QR$ comes from Gram–Schmidt, then R is invertible. | T / F |
| 16. The least-squares solution of $A\mathbf{x} = \mathbf{b}$ is always unique. | T / F |
| 17. Adding an x^2 column to a line fit (so fitting a parabola) can never increase the sum of squared errors. | T / F |

18. If Q is a 5×3 matrix with orthonormal columns, then $QQ^T = I_5$.

T / F

Section 2 — Multiple Choice

48 points

Circle the letter of the best answer.

1. If λ is an eigenvalue of A , then the eigenvectors for λ , together with $\mathbf{0}$, form:

- (A) a single vector
- (B) the whole space $\mathbb{R}^n$
- (C) a subspace called the eigenspace of λ
- (D) a set that is not closed under addition

2. A real 2×2 matrix has characteristic polynomial $\lambda^2 + 4$. Then:

- (A) its eigenvalues are 2 and -2
- (B) it is singular
- (C) it has no real eigenvectors
- (D) it is the identity matrix

3. Which condition guarantees that a 4×4 matrix is diagonalizable?

- (A) it has four distinct eigenvalues
- (B) it has no zero entries
- (C) its determinant is nonzero
- (D) all of its eigenvalues are real

4. Which of these matrices is *not* diagonalizable?

- (A) $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$
- (B) $\begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$
- (C) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- (D) $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

5. Which of these stochastic matrices is regular?

- (A) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
- (B) $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
- (C) $\begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$
- (D) $\begin{bmatrix} 0.3 & 0.6 \\ 0.7 & 0.4 \end{bmatrix}$

6. On $\mathbb{R}^2$ let $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 3u_2v_2$. Which vector is orthogonal to $(3, 1)$?

- (A) $(1, 3)$
- (B) $(1, -1)$
- (C) $(-1, 3)$
- (D) $(3, 1)$

7. In an inner product space, $\|\mathbf{u}\| = 2$ and $\|\mathbf{v}\| = 3$. The largest possible value of $|\langle \mathbf{u}, \mathbf{v} \rangle|$ is:
- (A) 5
 - (B) 1
 - (C) 9
 - (D) 6
8. If W is a plane through the origin in $\mathbb{R}^3$ (with the dot product), then $W^\perp$ is:
- (A) a plane through the origin
 - (B) a line through the origin
 - (C) just the origin
 - (D) all of $\mathbb{R}^3$
9. In Gram–Schmidt with $\mathbf{u}_1 = \mathbf{v}_1$, the next orthogonal vector is $\mathbf{u}_2 =$
- (A) $\mathbf{v}_2 - \frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1$
 - (B) $\mathbf{v}_2 - \frac{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle}{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle} \mathbf{u}_1$
 - (C) $\mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{v}_2$
 - (D) $\frac{\langle \mathbf{v}_2, \mathbf{u}_1 \rangle}{\langle \mathbf{u}_1, \mathbf{u}_1 \rangle} \mathbf{u}_1$
10. If $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an *orthonormal* basis of an inner product space, the coordinate of $\mathbf{v}$ along $\mathbf{u}_2$ is:
- (A) $\langle \mathbf{v}, \mathbf{u}_2 \rangle$
 - (B) $\langle \mathbf{u}_2, \mathbf{u}_2 \rangle$
 - (C) $\|\mathbf{v}\|$
 - (D) $\langle \mathbf{v}, \mathbf{v} \rangle$
11. The normal equations $A^T A \mathbf{x} = A^T \mathbf{b}$ always:
- (A) have no solution when $A \mathbf{x} = \mathbf{b}$ is inconsistent
 - (B) have at least one solution
 - (C) have exactly one solution
 - (D) have only the solution $\mathbf{x} = \mathbf{0}$
12. If the columns of A are linearly dependent, then the least-squares solutions of $A \mathbf{x} = \mathbf{b}$:
- (A) do not exist
 - (B) are unique
 - (C) are infinitely many, all with the same $A \hat{\mathbf{x}}$
 - (D) equal the exact solutions

13. To fit $y = a + bx$ to the points $(1, 3), (2, 5), (4, 9)$, the matrix A and vector $\mathbf{b}$ are:

- (A) $A = \begin{bmatrix} 1 & 3 \\ 1 & 5 \\ 1 & 9 \end{bmatrix}$, $\mathbf{b} = (1, 2, 4)$
- (B) $A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 4 & 1 \end{bmatrix}$, $\mathbf{b} = (3, 5, 9)$
- (C) $A = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 4 \end{bmatrix}$, $\mathbf{b} = (3, 5, 9)$
- (D) $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \end{bmatrix}$, $\mathbf{b} = (3, 5, 9)$

14. The least-squares line is the line that minimizes:

- (A) the sum of the squared vertical distances to the points
- (B) the sum of the absolute vertical distances
- (C) the largest vertical distance
- (D) the sum of the squared perpendicular distances

15. If Q is an orthogonal matrix, then Q^{-1} equals:

- (A) Q
- (B) $-Q$
- (C) Q^T
- (D) Q^2

16. Let Q be 5×3 with orthonormal columns. Which statement is true?

- (A) $QQ^T = I_5$
- (B) $Q^TQ = I_3$
- (C) Q is an orthogonal matrix
- (D) $\det Q = 1$

Section 3 — Short Answer / Definitions

32 points

Fill in each blank with the best term.

1. The polynomial $\det(A - \lambda I)$ is called the _____ of A .
2. The set of all eigenvectors for λ , together with $\mathbf{0}$, is called the _____ of λ .
3. The number of times λ appears as a root of the characteristic polynomial is its _____ multiplicity.
4. The dimension of the eigenspace of λ is its _____ multiplicity.
5. A matrix A is _____ if $A = PDP^{-1}$ for some invertible P and diagonal D .
6. For a Markov chain with transition matrix P , a probability vector $\mathbf{q}$ with $P\mathbf{q} = \mathbf{q}$ is called the _____ vector.
7. In an inner product space, the _____ of $\mathbf{v}$ is $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

8. The _____ between $\mathbf{u}$ and $\mathbf{v}$ is $d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$.
9. The inequality $|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ is called the _____ inequality.
10. The algorithm that turns a linearly independent set into an orthogonal set spanning the same subspace is called the _____ process.
11. A basis whose vectors are pairwise orthogonal is called an _____ basis.
12. The equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ are called the _____ equations.
13. The vector $\mathbf{b} - A \hat{\mathbf{x}}$ is called the _____ vector.
14. In a line fit, the matrix whose columns are the constant column $\mathbf{1}$ and the x -values is called the _____ matrix.
15. A square matrix Q with $Q^T Q = I$ is called an _____ matrix.
16. Writing $A = QR$ with Q having orthonormal columns and R upper triangular is called the _____ decomposition.

Section 4 — Quick Computations

24 points

No software needed for this section — these should be fast. Show minimal work.

1. Find all eigenvalues of $\begin{bmatrix} -1 & 0 & 0 \\ 4 & 6 & 0 \\ 2 & 3 & -5 \end{bmatrix}$.

Answer: _____

2. Is $\mathbf{v} = (1, 1)$ an eigenvector of $A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}$? If so, find its eigenvalue.

Answer: _____

3. A 3×3 matrix has the eigenvalue 5 with algebraic multiplicity 3 and an eigenspace of dimension 2. Is the matrix diagonalizable? Why?

Answer: _____

4. On $\mathbb{R}^2$ let $\langle \mathbf{u}, \mathbf{v} \rangle = u_1 v_1 + 4u_2 v_2$. Find the cosine of the angle between $\mathbf{u} = (1, 0)$ and $\mathbf{v} = (1, 1)$.

Answer: _____

5. In M_{22} with $\langle U, V \rangle = \text{tr}(U^T V)$ (the sum of products of corresponding entries), find $\langle A, B \rangle$ and $\|A\|$ for $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix}$.

Answer: _____

6. On $C[0, 1]$ with $\langle f, g \rangle = \int_0^1 fg \, dx$, are $f(x) = x$ and $g(x) = x - \frac{2}{3}$ orthogonal?

Answer: _____

7. Find the least-squares solution $\hat{x}$ of $\begin{bmatrix} 1 \\ 2 \end{bmatrix} x = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ and the length of the residual.

Answer: _____

8. The first two columns of a 3×3 orthogonal matrix are $(\frac{2}{3}, \frac{1}{3}, \frac{2}{3})$ and $(\frac{1}{3}, \frac{2}{3}, -\frac{2}{3})$. Find a possible third column.

Answer: _____

Section 5 — Inner Product or Not?**12 points**

(a) (6 points)

On P_2 (polynomials of degree at most 2) define $\langle p, q \rangle = p(0)q(0) + p(1)q(1)$. Is this an inner product? Justify: if it fails, name the axiom and give a specific counterexample; if it holds, verify all the axioms.

(b) (6 points)

On $\mathbb{R}^2$ define $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + u_1v_2 + u_2v_1 + 2u_2v_2$. Is this an inner product? Justify: if it fails, name the axiom and give a counterexample; if it holds, verify all the axioms.

Section 6 — Computation with Software**48 points**

You may use a row-reduction calculator or software for this section. Show the matrix or system you set up and row reduced, and clearly state your final answer. Box or circle your final answers.

Problem 1 — Diagonalization**(6 points)**

Let $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & 0 \\ -3 & 1 & -1 \end{bmatrix}$. **(a)** Find the characteristic polynomial and eigenvalues (2 pts). **(b)** Find an eigenvector for each eigenvalue (2 pts). **(c)** Give P and D with $A = PDP^{-1}$ (2 pts).

Work / setup (show the matrix or system you row reduced)**Final answer****Problem 2 — A Defective Matrix****(6 points)**

Let $B = \begin{bmatrix} 3 & 1 & 2 \\ 0 & 3 & -1 \\ 0 & 0 & 5 \end{bmatrix}$. **(a)** Find the eigenvalues and their algebraic multiplicities (2 pts). **(b)** Find a basis for the eigenspace of $\lambda = 3$ (2 pts). **(c)** Is B diagonalizable? Explain (2 pts).

Work / setup (show the matrix or system you row reduced)**Final answer**

Problem 3 — Weighted Gram–Schmidt**(6 points)**

On $\mathbb{R}^3$ let $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + 3u_2v_2 + 2u_3v_3$, $\mathbf{v}_1 = (1, 0, 1)$, $\mathbf{v}_2 = (2, 1, 0)$. **(a)** Use Gram–Schmidt to find an orthogonal pair $\mathbf{u}_1, \mathbf{u}_2$ (4 pts). **(b)** Find the orthonormal pair $\mathbf{q}_1, \mathbf{q}_2$ (2 pts).

Work / setup (show the matrix or system you row reduced)**Final answer****Problem 4 — Gram–Schmidt and QR****(6 points)**

Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 3 \\ 2 & 0 \end{bmatrix}$. **(a)** Use Gram–Schmidt with the dot product on the columns of A to find orthonormal $\mathbf{q}_1, \mathbf{q}_2$ (3 pts). **(b)** Find $R = Q^T A$ so that $A = QR$ (2 pts). **(c)** Check that $Q^T Q = I$ (1 pt).

Work / setup (show the matrix or system you row reduced)**Final answer**

Problem 5 — Best-Fit Line**(6 points)**

Find the least-squares line $y = a + bx$ through $(1, 6), (2, 5), (3, 3), (4, 2), (5, 1)$. **(a)** Compute $A^T A$ and $A^T \mathbf{b}$ (2 pts). **(b)** Solve for a, b (2 pts). **(c)** Predict y at $x = 6$ (1 pt), and say what the slope tells you (1 pt).

Work / setup (show the matrix or system you row reduced)**Final answer****Problem 6 — Best-Fit Parabola****(6 points)**

Fit $y = c_0 + c_1x + c_2x^2$ by least squares to $(-2, 0), (-1, 1), (0, -1), (1, -1), (2, 6)$. **(a)** Write A (1 pt). **(b)** Compute $A^T A$ and $A^T \mathbf{b}$ (2 pts). **(c)** Solve for c_0, c_1, c_2 (2 pts). **(d)** Predict y at $x = 3$ (1 pt).

Work / setup (show the matrix or system you row reduced)**Final answer**

Problem 7 — Markov Chain**(6 points)**

A Markov chain has transition matrix $P = \begin{bmatrix} 0.6 & 0.1 & 0.2 \\ 0.3 & 0.7 & 0.2 \\ 0.1 & 0.2 & 0.6 \end{bmatrix}$ and initial state $\mathbf{x}_0 = (0.5, 0.5, 0)$. **(a)** Is P stochastic? Why? (1 pt) **(b)** Find $\mathbf{x}_1$ and $\mathbf{x}_2$ (2 pts). **(c)** Find the steady-state vector $\mathbf{q}$ (3 pts).

Work / setup (show the matrix or system you row reduced)**Final answer****Problem 8 — An Orthogonal Complement****(6 points)**

Let $W = \text{span}\{(1, 1, 0, 1), (0, 1, 1, -1)\} \subseteq \mathbb{R}^4$ with the dot product. **(a)** Set up the system whose solutions are the vectors in $W^\perp$ (2 pts). **(b)** Row reduce it and find a basis for $W^\perp$ (3 pts). **(c)** What is $\dim W^\perp$? (1 pt)

Work / setup (show the matrix or system you row reduced)**Final answer**